

Chapter 6

Dominance and Best Response

The simplest, most compelling theory of behavior is that rational players never play a strategy that does strictly worse than some alternative, no matter what everyone else does.

This chapter makes that idea precise, then asks what a player should do once dominance alone cannot settle the question. We will meet the two concepts on which every later solution concept is built.

What Is Dominance?

	L	R
U	2, 3	5, 0
D	1, 0	4, 3

Suppose we are player 1. Player 2 picks L or R on her own; a good way to think about it is that she has already chosen, and we simply have not seen which.

Whatever she chose, U beats D : $2 > 1$ against L , and $5 > 4$ against R . We say D is **dominated** by U . No rational player should ever play a dominated strategy.

Neither of player 2's strategies is dominated here, so dominance alone says nothing about her choice.

A Tie Is Not Enough

	L	C	R
U	8, 3	0, 4	4, 4
M	4, 2	1, 5	5, 3
D	3, 7	0, 1	2, 0

Here D is dominated by M : $4 > 3$, $1 > 0$, $5 > 2$. But D is *not* dominated by U , since against C both give player 1 exactly 0. A tie is not enough.

The same goes for player 2: C beats R against M and D , but they tie against U , so C does not dominate R .

Dominance requires a strictly higher payoff against *every* strategy of the others. To stress this, it is sometimes called **strict dominance**.

Example: A Mixed Strategy Can Dominate

	<i>L</i>	<i>R</i>
<i>U</i>	4, 1	0, 2
<i>M</i>	0, 0	4, 0
<i>D</i>	1, 3	1, 2

No pure strategy dominates *D*: *U* beats *D* against *L* but loses against *R*, and *M* is the mirror image.

The mixed strategy $\sigma_1 = (\frac{1}{2}, \frac{1}{2}, 0)$ does the job:

$$u_1(\sigma_1, L) = \frac{1}{2}(4) + \frac{1}{2}(0) = 2 > 1 \quad \text{and} \quad u_1(\sigma_1, R) = \frac{1}{2}(0) + \frac{1}{2}(4) = 2 > 1.$$

Averaging *U* and *M* beats *D* everywhere, even though neither alone does.

Alternating large and small payoffs across rows are the sign to look for.

The Formal Definition

Definition. Player i 's pure strategy s_i is **dominated** if some strategy σ_i (pure or mixed) satisfies $u_i(\sigma_i, s_{-i}) > u_i(s_i, s_{-i})$ for every strategy profile s_{-i} of the other players. We say σ_i **dominates** s_i .

A pure strategy that dominates every other pure strategy is called a **dominant strategy**.

To check whether s_i is dominated: first look for a dominating pure strategy, comparing rows for player 1 or columns for player 2, holding the opponent's strategy fixed. Only if that fails is it worth searching for a dominating mixture. One dominating strategy is enough; there is no need to find them all.

Discussion: Find the Dominated Strategies

In each game, which strategies are dominated, and by what?

	L	R
A	3, 3	2, 0
B	4, 1	8, -1

(a)

	L	C	R
U	5, 9	0, 1	4, 3
M	3, 2	0, 9	1, 1
D	2, 8	0, 1	8, 4

(b)

Remember to check each player's *own* payoffs, and to look for mixtures once pure strategies fail.

Answer: Find the Dominated Strategies

Game (a). For player 1, A is dominated by B : $4 > 3$ and $8 > 2$. For player 2, R is dominated by L : $3 > 0$ and $1 > -1$. Both players have a dominant strategy, and the game will be played at (B, L) .

Game (b). For player 2, R is dominated by L : $9 > 3$, $2 > 1$, $8 > 4$. C is not dominated: it beats L against M .

For player 1, nothing is dominated. U and D are each the unique best reply to some column (L and R). M looks weak, but every strategy of player 1 pays exactly 0 against C , so no pure strategy or mixture can beat M *strictly* there. The tie in the C column protects M .

Revisiting the Prisoners' Dilemma

	Cooperate	Defect
Cooperate	2, 2	0, 3
Defect	3, 0	1, 1

For each player, Cooperate is dominated by Defect. Rational, self-interested players are therefore driven to (1, 1), even though both prefer (2, 2).

The strategic tension here is the clash between individual and group interests. The players could agree out loud to cooperate, but once they choose separately, neither has any incentive to follow through.

If fear of later retribution would stop a player from defecting, that fear belongs in the payoffs, and the game is no longer a Prisoners' Dilemma.

The Concept of Efficiency

Definition. Pure strategy profile s' is **more efficient** than s if $u_i(s') \geq u_i(s)$ for every player i , with strict inequality for at least one. A profile is **efficient** (or **Pareto efficient**) if no other pure profile is more efficient than it. Here we compare matrix cells, not randomized outcomes.

In the Prisoners' Dilemma, $(2, 2)$ is more efficient than $(1, 1)$. The efficient profiles are $(2, 2)$, $(0, 3)$, and $(3, 0)$: only $(1, 1)$ fails.

Dominance is about what an individual should do; efficiency is about what the group could achieve together. The tension between them is what makes the Prisoners' Dilemma worth studying.

Discussion: Which Profiles Are Efficient?

	L	C	R
U	8, 3	0, 4	4, 4
M	4, 2	1, 5	5, 3
D	3, 7	0, 1	2, 0

Return to the game from the tie example. Which of its nine pure strategy profiles are efficient among the matrix cells?

For each profile you rule out, name a profile that is more efficient than it.

Answer: Which Profiles Are Efficient?

Among pure profiles, three are efficient: (U, L) with payoffs $(8, 3)$, (U, R) with $(4, 4)$, and (D, L) with $(3, 7)$.

Nobody else gives player 1 at least 8, or player 2 at least 7, so (U, L) and (D, L) cannot be improved on. For (U, R) , an improvement would need at least 4 for both players, and no other cell offers that.

Each other profile is beaten by one of these three: $(8, 3)$ improves on $(4, 2)$, $(5, 3)$, and $(2, 0)$; $(4, 4)$ improves on $(0, 4)$ and $(0, 1)$; $(3, 7)$ improves on $(1, 5)$.

Notice that efficiency says nothing about what rational players will *do*. D is dominated, so (D, L) is efficient yet will never be played.

Best Response

Matching pennies, the battle of the sexes, and the coordination games all have *no* dominated strategies. Dominance alone cannot tell us how to play them.

Definition. Given belief θ_{-i} , player i 's pure strategy s_i is a **best response** if $u_i(s_i, \theta_{-i}) \geq u_i(s'_i, \theta_{-i})$ for every alternative $s'_i \in S_i$. We write $BR_i(\theta_{-i})$ for the set of best responses.

A rational player forms a belief about the others, then plays a strategy that maximizes her expected payoff against it. In a finite game every belief has at least one best response, and $BR_i(\theta_{-i})$ is always written as a set, even with a single element.

Example: Computing Best Responses

	<i>L</i>	<i>C</i>	<i>R</i>
<i>U</i>	2, 6	0, 4	4, 4
<i>M</i>	3, 3	0, 0	1, 5
<i>D</i>	1, 1	3, 5	2, 3

Suppose player 1 believes $\theta_2 = (\frac{1}{3}, \frac{1}{2}, \frac{1}{6})$. Her expected payoffs are

$$\begin{aligned} U &: \frac{1}{3}(2) + \frac{1}{2}(0) + \frac{1}{6}(4) = \frac{8}{6}, & M &: \frac{1}{3}(3) + \frac{1}{2}(0) + \frac{1}{6}(1) = \frac{7}{6}, \\ D &: \frac{1}{3}(1) + \frac{1}{2}(3) + \frac{1}{6}(2) = \frac{13}{6}. \end{aligned}$$

Since D pays the most, $BR_1(\frac{1}{3}, \frac{1}{2}, \frac{1}{6}) = \{D\}$, a single best response.

A Best Response Can Tie

Using the same matrix, suppose player 2 believes $\theta_1 = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$: probability $\frac{1}{2}$ on U and the rest split evenly. His expected payoffs are

$$\begin{aligned} L : \frac{1}{2}(6) + \frac{1}{4}(3) + \frac{1}{4}(1) &= 4, & C : \frac{1}{2}(4) + \frac{1}{4}(0) + \frac{1}{4}(5) &= \frac{13}{4}, \\ R : \frac{1}{2}(4) + \frac{1}{4}(5) + \frac{1}{4}(3) &= 4. \end{aligned}$$

Both L and R tie for the highest payoff, so $BR_2(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}) = \{L, R\}$: a best-response set can contain more than one strategy.

Discussion: More Best Responses

	<i>L</i>	<i>C</i>	<i>R</i>
<i>U</i>	2, 6	0, 4	4, 4
<i>M</i>	3, 3	0, 0	1, 5
<i>D</i>	1, 1	3, 5	2, 3

For the same game, find the following best-response sets.

1. $BR_1(\theta_2)$ for $\theta_2 = (\frac{1}{6}, \frac{1}{3}, \frac{1}{2})$.
2. $BR_1(\theta_2)$ for $\theta_2 = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.
3. $BR_2(\theta_1)$ for $\theta_1 = (\frac{1}{2}, \frac{1}{2}, 0)$.

Answer: More Best Responses

1. U pays $\frac{1}{6}(2) + \frac{1}{3}(0) + \frac{1}{2}(4) = \frac{14}{6}$; M pays $\frac{1}{6}(3) + \frac{1}{3}(0) + \frac{1}{2}(1) = 1$; D pays $\frac{1}{6}(1) + \frac{1}{3}(3) + \frac{1}{2}(2) = \frac{13}{6}$. So $BR_1 = \{U\}$, by a margin of one sixth.
2. U pays $\frac{1}{3}(2+0+4) = 2$; M pays $\frac{1}{3}(3+0+1) = \frac{4}{3}$; D pays $\frac{1}{3}(1+3+2) = 2$.
So $BR_1 = \{U, D\}$.
3. L pays $\frac{1}{2}(6) + \frac{1}{2}(3) = \frac{9}{2}$; C pays $\frac{1}{2}(4) + \frac{1}{2}(0) = 2$; R pays $\frac{1}{2}(4) + \frac{1}{2}(5) = \frac{9}{2}$.
So $BR_2 = \{L, R\}$.

Small changes in belief can change the best response, or create a tie. Every strategy in this game is a best response to *some* belief; we return to this observation shortly. A mixture over tied best responses is also optimal; BR_i here lists only pure strategies.

Discussion: Taking Your Friend's Advice

	A	B
A	1, 3	0, 0
B	0, 0	3, 1

You are player 2 in this battle-of-the-sexes game. Knowing player 1 well, you are sure he will choose B , so you plan to play B too.

A friend, over bubble tea, scolds you for “giving in.” Demand the best outcome for yourself, he says: play A .

Should you take the advice?

Answer: Taking Your Friend's Advice

No. Your choice has no effect on player 1's decision, since strategies are chosen simultaneously and independently. Playing A against a player who chooses B earns you 0 instead of 1.

The rules of the game give you no way to signal your intentions through your choice. If signaling were possible, it would have to be written into the game, and the game would be a different one.

You control your own action. If your friend has new information about player 1, use it to revise your belief. Wanting a better outcome is not itself a reason to change that belief; given it, choose a best response.

Playing a best response is a computation. Forming the belief is where the real difficulty lies.

Dominance and Best Response Compared

Let UD_i be the set of player i 's pure strategies that are not dominated, and B_i the set of pure strategies that are a best response to *some* belief:

$$B_i = \{s_i \in S_i : s_i \in BR_i(\theta_{-i}) \text{ for some belief } \theta_{-i}\}.$$

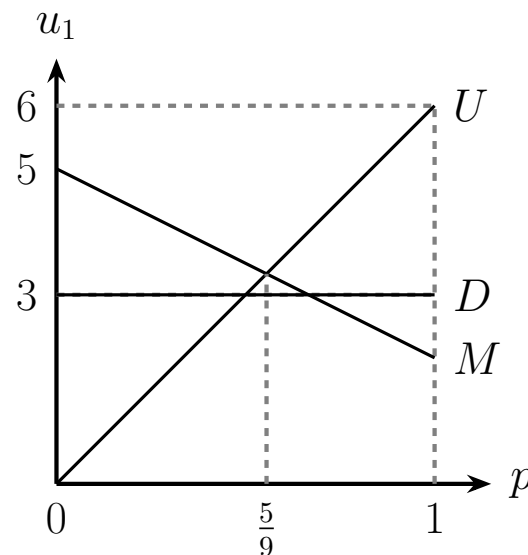
Finding B_i directly requires searching over beliefs. Dominance offers another route through payoff comparisons, including mixtures.

Result. In a finite two-player game, $B_1 = UD_1$ and $B_2 = UD_2$.

A dominating strategy wins against every opposing strategy, hence under every belief. The converse is the useful part: an undominated strategy is a best response to *some* belief.

Visualizing Best Response

	L (p)	R ($1 - p$)
U	6, 3	0, 1
M	2, 1	5, 0
D	3, 2	3, 1



For player 2, L dominates R , so $UD_2 = B_2 = \{L\}$. Player 1's expected payoffs, with $p = \Pr(L)$, are $6p$ for U , $5 - 3p$ for M , and 3 for D .

U is best for $p > \frac{5}{9}$, M for $p < \frac{5}{9}$, and both tie at $p = \frac{5}{9}$: $B_1 = \{U, M\}$. The mixture $(\frac{1}{3}, \frac{2}{3}, 0)$ dominates D , paying $\frac{10}{3} > 3$ against both columns. The equal mixture fails: it pays only $\frac{5}{2}$ against R .

A Procedure for Computing $B_i = UD_i$

For a two-player matrix game, the result gives a practical recipe.

1. Find the best responses to the simplest beliefs, those that put all probability on a single strategy of the opponent. Each of these is in B_i , hence in UD_i .
2. Find the strategies dominated by another pure strategy. These are out of UD_i , hence out of B_i .
3. Test each remaining strategy for dominance by a mixture. This is the hard step, but with luck it is rarely needed.

Discussion: Running the Procedure

	L	R
W	2, 4	2, 5
X	2, 0	7, 1
Y	6, 5	1, 2
Z	5, 6	3, 0

Compute $B_1 = UD_1$ for player 1 using the three-step procedure.

For any strategy you conclude is undominated but is not a best response to a pure belief, name a belief to which it *is* a best response.

Answer: Running the Procedure

Step 1. Against L , player 1's best response is Y (6); against R it is X (7). So $X, Y \in B_1$.

Step 2. W is dominated by Z : $5 > 2$ and $3 > 2$. So $W \notin B_1$.

Step 3. Only Z is undecided. It is enough to test mixtures of X and Y : replacing any weight on W with Z improves payoffs, and removing weight on Z preserves strict dominance over Z . Put weight p on X and $1 - p$ on Y . Against L it must beat 5: $2p + 6(1 - p) > 5$, so $p < \frac{1}{4}$. Against R it must beat 3: $7p + (1 - p) > 3$, so $p > \frac{1}{3}$. No p satisfies both, so Z is undominated.

Hence $B_1 = UD_1 = \{X, Y, Z\}$. Indeed Z is the best response to $\theta_2 = (\frac{3}{5}, \frac{2}{5})$: it pays $\frac{21}{5}$, while X and Y each pay 4 and W pays 2.

Weak Dominance

Definition. Strategy σ_i (pure or mixed) **weakly dominates** pure strategy s_i if $u_i(\sigma_i, s_{-i}) \geq u_i(s_i, s_{-i})$ for every s_{-i} , with strict inequality for at least one s_{-i} .

	L	M
X	3, 5	3, 5
Y	7, 0	1, 1

For player 2, M ties L against X but strictly beats it against Y : M weakly dominates L .

Weak dominance encodes caution: if player 2 puts any positive probability on Y , M is her only best response. But if she is *certain* of X , choosing L is perfectly rational, so we work with strict dominance from here on.

Discussion: Two Delivery Apps

Two food-delivery apps simultaneously set their restaurant commission rate, Low or High.

- If both set Low, each earns 2 (thin margins, but both keep their restaurants).
- If both set High, each earns 1 (restaurants grumble, but stay for lack of a better option).
- If one sets Low while the other sets High, the Low app earns 4 by poaching restaurants, and the High app earns 0.

Write the matrix. Is either strategy dominated? What does that imply about how this game will be played, and is the outcome efficient?

Answer: Two Delivery Apps

	Low	High
Low	2, 2	4, 0
High	0, 4	1, 1

For each app, Low dominates High: against Low, $2 > 0$; against High, $4 > 1$. Both apps rationally set commissions Low.

Unlike the Prisoners' Dilemma, this is not a tragic outcome: $(2, 2)$ is efficient, since the only profile that gives either app more, $(4, 0)$ or $(0, 4)$, gives the other less. Individual and group incentives happen to agree here.

Chapter in a Nutshell

- A dominated strategy does strictly worse than some alternative, pure or mixed, against every possible play of the others; no rational player uses one.
- Efficiency asks what is good for the group; dominance asks what is good for the individual, and the two can conflict, as in the Prisoners' Dilemma.
- A best response maximizes expected payoff against a specific belief; in a finite two-player game a strategy is undominated exactly when it is a best response to some belief.
- Weak dominance is a cautious refinement, but a weakly dominated strategy can still be a best response to some belief.

Key Concepts

- Dominated strategy, dominant strategy, and weak dominance describe when one plan beats another, strictly in every case, or weakly in every case with at least one strict gain.
- Efficiency (Pareto efficiency) asks whether some other outcome could make everyone at least as well off and someone strictly better off.
- Best response and best-response set $BR_i(\theta_{-i})$ describe optimal play against a specific belief.
- The result $B_i = UD_i$ connects dominance to best response: undominated is the same as justified by some belief.